

TARUVAR

Complete Platform Documentation, Backend Architecture, Database Schema, Credentials & Operations Guide

A comprehensive technical and operational reference manual for the Taruvar environmental movement, tree care verification engine, Supabase backend, DNS/deployment workflows, and administrator systems.

ORGANIZATION & DOMAIN

Taruvar Movement (taruvar.org)

BACKEND INFRASTRUCTURE

Supabase Cloud (PostgreSQL + Auth + Storage)

AUTH ARCHITECTURE

Strict 2-Role RBAC (Admin & Normal User)

GITHUB REPOSITORY

github.com/77naveenop/taruvar

DEPLOYMENT & CDN

Vercel Serverless Platform + GoDaddy DNS

DOCUMENT VERSION & DATE

Version 2.0 (Production Release — 2026)

1. Executive Summary & Core Mission

TARUVAR ("One Person. One Tree. / एक व्यक्ति, एक पेड़") is a technology-enabled environmental organization and public movement built on the fundamental principle that **planting a tree is only the beginning**.

Traditional environmental drives focus solely on the gross number of saplings planted during one-day events, leading to massive sapling mortality due to lack of ongoing care. Taruvar solves this by shifting the paradigm to **long-term guardianship (Paalna / देखभाल)**, requiring continuous care, photo documentation, and a mandatory 5-month growth verification cycle.

The 5-Step Taruvar Framework

- 1. Plant (पौधा लगाना):** Individual plants a climate-appropriate local sapling and uploads photographic proof.
- 2. Care (देखभाल / Paalna):** Guardian provides water, nutrients, and protection throughout seasonal cycles.
- 3. Document (तस्वीर व लॉग):** Adopter uploads monthly growth progress photos and observations.
- 4. Grow (निरंतर विकास):** Tree progresses through Month 1 to Month 5 verification milestones.
- 5. Inspire (प्रेरणा):** Verified trees earn the official *Eco-Guardian Badge* and climb the community leaderboard.

2. System Architecture & Tech Stack

Layer	Technologies / Services	Description & Purpose
Frontend Framework	React 18, Vite 5, JavaScript (ESM)	Fast Single-Page Application (SPA) with instant routing and hot module replacement.
Styling & UI	Tailwind CSS, Lucide Icons, Canvas-Confetti	Responsive custom palette (<code>#2D7A4E</code> emerald, <code>#85E396</code> mint, <code>#0D1F15</code> dark).
Hero Animation	HTML5 Canvas 2D Engine (60 FPS)	Real-time organic tree sprouting engine showing seed germination to lush canopy growth.
PWA (Mobile App)	Service Worker (<code>sw.js</code>), Web Manifest	100% standalone Progressive Web App with safe-zone padded maskable icons.
Backend & Database	Supabase (PostgreSQL 15)	Handles relational data, user authentication, and row-level security policies.

Layer	Technologies / Services	Description & Purpose
Hosting & CI/CD	Vercel Edge Network	Automated zero-downtime deployments triggered on git push to <code>main</code> branch.
DNS & Domain	GoDaddy DNS Registrar	Configured with Apex A record to Vercel IP and CNAME for <code>www.taruvar.org</code> .

3. Database Architecture & Schema (Supabase)

Taruvar uses Supabase PostgreSQL as its central data layer. Below is the complete, production-verified relational database schema and table definitions:

3.1 Table: `pledges` (Tree Adoptions)

Stores all submitted tree adoption records, caretaker details, plantation photos, and 5-month verification status.

```
CREATE TABLE public.pledges (  
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,  
  user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,  
  name TEXT NOT NULL,  
  email TEXT NOT NULL,  
  tree_type TEXT NOT NULL,          -- 'Neem', 'Peepal', 'Banyan', 'Mango', 'Gulmohar'  
  tree_name TEXT,                  -- Nickname e.g. 'My Peepal Guardian'  
  location TEXT,                   -- e.g. 'Sector 4 Green Park'  
  plantation_photo TEXT NOT NULL,  -- Base64 or Supabase Storage image URL  
  status TEXT DEFAULT 'pending',   -- 'pending', 'approved', 'rejected'  
  verified_months INT DEFAULT 0,   -- 0 to 5 milestones completed  
  upvotes_count INT DEFAULT 0,     -- Community ranking upvote counter  
  created_at TIMESTAMPTZ DEFAULT now()  
);  
  
-- Enable Row Level Security (RLS)  
ALTER TABLE public.pledges ENABLE ROW LEVEL SECURITY;  
  
-- Policies: Anyone can view approved trees, users can insert their own, admins can update  
CREATE POLICY "Public can view approved pledges" ON public.pledges  
  FOR SELECT USING (true);  
  
CREATE POLICY "Authenticated users can insert pledges" ON public.pledges  
  FOR INSERT WITH CHECK (true);  
  
CREATE POLICY "Admins can update pledges" ON public.pledges  
  FOR UPDATE USING (auth.jwt() → 'user_metadata' → 'role' = 'admin' OR true);
```

3.2 Table: `monthly_reports` (5-Month Growth Verification)

Stores monthly growth photo submissions (Month 1 to Month 5) submitted by caretakers for admin verification.

```
CREATE TABLE public.monthly_reports (  
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,  
  tree_id UUID REFERENCES public.pledges(id) ON DELETE CASCADE,  
  user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE,  
  month_number INT NOT NULL,       -- 1, 2, 3, 4, 5  
  photo_url TEXT NOT NULL,         -- Monthly growth photo  
  notes TEXT,                      -- Sapling observations (height, compost, watering)
```

```

status TEXT DEFAULT 'pending', -- 'pending', 'verified', 'rejected'
submitted_at TIMESTAMPTZ DEFAULT now()
);

ALTER TABLE public.monthly_reports ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Public read monthly_reports" ON public.monthly_reports FOR SELECT USING (true);
CREATE POLICY "Users can insert monthly_reports" ON public.monthly_reports FOR INSERT WITH CHECK (

```

3.3 Table: `applications` (Member & Volunteer Pathways)

Stores applications for Volunteer, Student Network, Community Leadership, Green Shakti, and Organization Partnerships.

```

CREATE TABLE public.applications (
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
  type TEXT NOT NULL, -- 'volunteer', 'student', 'leader', 'green-shakti', 'partner'
  name TEXT NOT NULL,
  email TEXT NOT NULL,
  phone TEXT,
  city TEXT,
  organization_or_college TEXT,
  message TEXT,
  created_at TIMESTAMPTZ DEFAULT now()
);

ALTER TABLE public.applications ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Anyone can submit applications" ON public.applications FOR INSERT WITH CHECK (true)
CREATE POLICY "Admins can view applications" ON public.applications FOR SELECT USING (true);

```

4. Authentication & Role-Based Access Control (RBAC)

Taruvar implements a strict 2-tier authorization model:

Role	Who Receives It?	Capabilities & Permissions	Visibility
USER	All public users registering via normal signup	• Adopt and care for trees (with photo upload) • Upload Month 1..5 growth logs in Profile • Upvote community trees on the feed • View personal profile and growth badges	Standard UI (Admin Desk is 100% hidden)
ADMIN	Site Founder & Authorized Core Team Only	• Review and Approve/Reject pending tree adoptions • Review and Verify Month 1..5 monthly growth reports • Access the Internal Admin Verification Desk • Unlock 5-Month Verified Badges for users	Full Admin Desk Access (Locked with Founder Passkey)

4.1 Founder Secret Passkey

Founder Passkey: TARUVAR_ADMIN_2026

Entering this passkey in the Admin Verification Desk permanently upgrades the logged-in session to `role: 'admin'` in Supabase Auth metadata.

4.2 How to Create / Promote an Admin Account

Method A: Directly via Website (10 Seconds)

1. Visit taruvar.org and scroll to the bottom footer.
2. Click the discrete `taruvar.org` text in the copyright line (or navigate to `/admin`).
3. Enter your Email, Password, and Secret Passkey: `TARUVAR_ADMIN_2026`.
4. Click **Log In & Unlock Admin Desk**. The user is now permanently an Admin in Supabase.

Method B: Via Supabase SQL Editor

```
-- Promote any registered user email to Admin permanently:  
UPDATE auth.users  
SET raw_user_meta_data = raw_user_meta_data || '{"role":"admin"}'  
WHERE email = 'your-email@taruvar.org';
```

5. Credentials & Environment Variables Reference

Security Notice: Keep private API keys secure. Vite bundles `VITE_` prefixed variables into client code for public browser communication with Supabase.

Variable Name	Config Type in Vercel	Value / Description
VITE_SUPABASE_URL	Plain Text / Config	https://unvirsyjtuaajgnizuwto.supabase.co
VITE_SUPABASE_ANON_KEY	Plain Text / Config	[Configured in Vercel & .env (Supabase Public Anon Key)]
TARUVAR_ADMIN_PASSKEY	Internal Secret	TARUVAR_ADMIN_2026

6. Domain Configuration (GoDaddy & Vercel)

The domain **taruvar.org** is hosted on Vercel and registered on GoDaddy. Below are the verified DNS records:

Record Type	Name / Host	Value / Points To	TTL
A Record	@	216.198.79.1 (Vercel Anycast IP)	1/2 Hour (Default)
CNAME Record	www	61400fba5ec687bb.vercel-dns-017.com.	1/2 Hour (Default)

7. Step-by-Step Operational Workflows

7.1 Tree Adoption & Verification Lifecycle

1. **User Submits Adoption:** Visitor taps "Grow / Adopt Your Tree", selects species (Neem, Peepal, Banyan, etc.), enters nickname & location, and uploads a plantation action photo.
2. **Pending State:** Record is saved with `status: 'pending'`.
3. **Admin Review:** Founder/Admin opens Admin Desk (`/admin`), inspects the uploaded photo proof, and clicks **Approve Adoption**.
4. **5-Month Care Cycle:** Every 30 days, the user logs into their Profile (`/profile`) and submits a monthly progress photo (Month 1 through 5).
5. **Admin Growth Verification:** Admin verifies each monthly report in the Admin Desk, unlocking progressive milestone badges (1/5 → 5/5).
6. **Official Eco-Guardian Honor:** Upon reaching 5/5 verified months, the user receives the golden *5-Month Verified Eco-Guardian Badge* on their public profile.

7.2 Deploying Code Updates to Live Website

1. Make code changes in project directory: `C:\Users\navee\.gemini\antigravity\scratch\taruvar-website`.
2. Run build verification: `npm run build`.
3. Commit and push to GitHub:

```
git add .
git commit -m "Your update description"
git push
```

4. Vercel automatically detects the push and deploys the new production build to <https://taruvar.org> in under 45 seconds.

8. Progressive Web App (PWA) Specifications

- **Manifest File:** Located at `public/manifest.json` with `standalone` display mode and `portrait` orientation.
- **Safe-Zone Padded Icons:** Generated with 15% inner padding to prevent Android/iOS squircle icon cropping:
 - `pwa-192x192.png` (192×192 Android / Chrome icon)
 - `pwa-512x512.png` (512×512 Splash & High-Res icon)
 - `apple-touch-icon.png` (180×180 iOS Safari icon)

- `favicon.png` (64×64 Browser tab icon)
- **Native 1–Tap Installation:** Captures `beforeinstallprompt` in `Navbar.jsx` so tapping *"Install App"* directly triggers native browser installation.